Exponential functions

1 Consider the function $y = 5 \times 4^x$. Find the value of y when:

a $x = 0$ **b** $x = 2$ **c** $x = -1$ **d** $x = 3$

2 Consider the function $f(x) = \left(\frac{4}{3}\right)^x$.

a Evaluate $f(0)$. **b** Evaluate $f(-2)$.

3 Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

4 Find the missing coordinate in each ordered pair that is a solution to $y = 5^x$:

a $(3, \square)$ **b** $(\square, 5)$ **c** $(-1, \square)$ **d** $\left(\square, \frac{1}{25}\right)$

5 Find the missing coordinate in each ordered pair that is a solution to $y = -2^x$:

a $(5, \square)$ **b** $\left(\square, -\frac{1}{8}\right)$ **c** $(-1, \square)$ **d** $(\square, -16)$

6 If the point $(3, m)$ lies on the curve $y = 2^x$, solve for m.

7 The points $(3, n)$, $(k, 16)$ and $\left(m, \frac{1}{4}\right)$ all lie on the curve with equation $y = 2^x$.

a Solve for n. **b** Solve for k. **c** Solve for m.

Characteristics of exponential functions

8 Do either of the functions $y = 9^x$ or $y = 9^{-x}$ have x-intercepts? Explain your answer.

9 For what values of x will 6^x have a negative value?

10 Determine the y-intercept of all exponential functions of the form:

a $y = a^x$ **b** $y = a^{-x}$ **c** $y = -a^x$ **d** $y = -a^{-x}$

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11 Consider an exponential curve with equation of the form $y = a^x$, where $a > 0$.

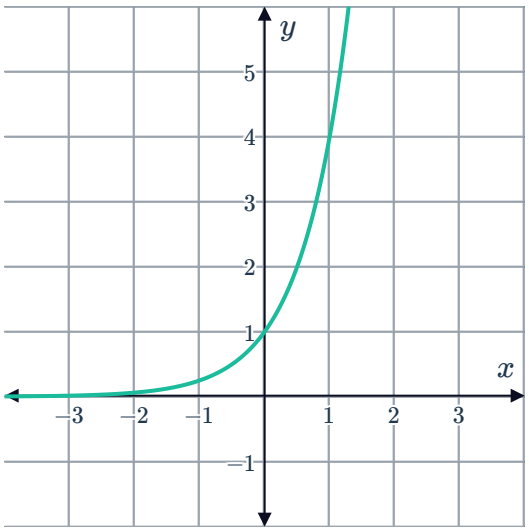
- **a** What is the y-value of the y-intercept of all exponential curves of this form?
- **b** What is the y-value of the y-intercept of $y = (3.5)^x$?

12 Consider the expression 3^x.

- **a** Evaluate the expression when $x = -4$.
- **b** Evaluate the expression when $x = 0$.
- **c** Evaluate the expression when $x = 4$.
- **d** What happens to the value of 3^x as x gets larger?
- **e** What happens to the value of 3^x as x gets smaller?

13 Consider the graph of the function $y = 4^x$:

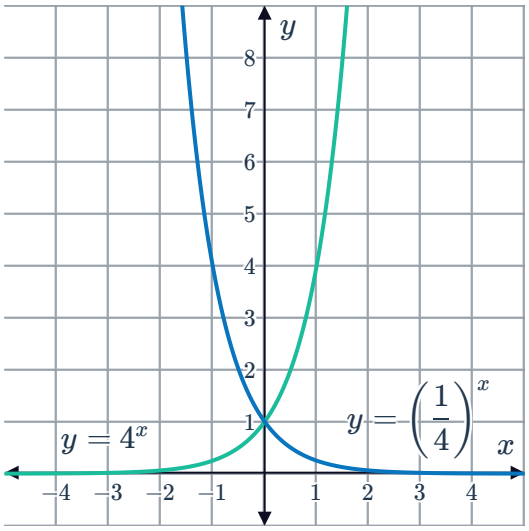
- **a** Is each y-value of the function positive or negative?
- **b** State the value of y the graph approaches but does not reach.
- **c** State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.



14 Given the function $y = -10^x$, what is the largest possible function value? Explain your answer.

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15 Consider the graphs of the functions $y = 4^x$ and $y = \left(\frac{1}{4}\right)^x$.



a Which function is an increasing function?

b How would you describe the rate of increase of $y = 4^x$?

16 Of the two functions $y = 4^x$ and $y = 5^x$, which increases more rapidly for $x > 0$?

17 Consider the two functions $y = 4^{-x}$ and $y = 5^{-x}$. Which function decreases more rapidly for $x > 0$?

18 Consider the function $y = 3^x$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c State the domain of the function.

d State the range of the function.

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19 Consider the function $y = 2^{-x}$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c Find the y-value of the y-intercept of the function.

d State the domain of the function.

e State the range of the function.

20 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

b Can the value of y ever be zero or positive? Explain your answer.

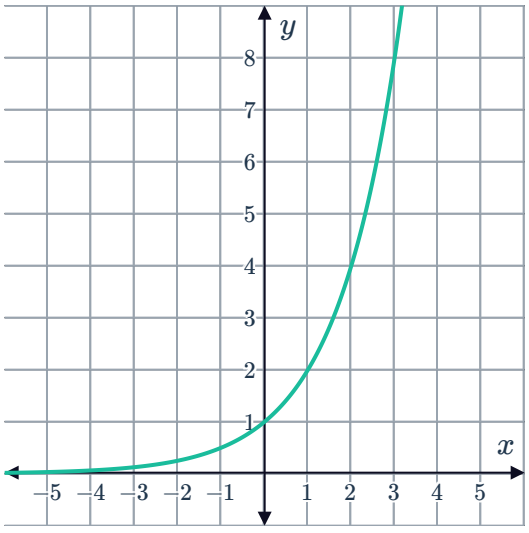
c Is $y = -(2^x)$ an increasing or decreasing function?

d Describe the behavior of the function as x increases.

e State the domain of the function.

f State the range of the function.

21 Consider the graph of the function $y = 2^x$:



a State the y-intercept of this graph.

b Does the graph have an x-intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.

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22 Consider the functions $y = 2^{-x}$, $y = 3^{-x}$ and $y = 5^{-x}$.

a State a characteristic that all the graphs of the functions have in common.

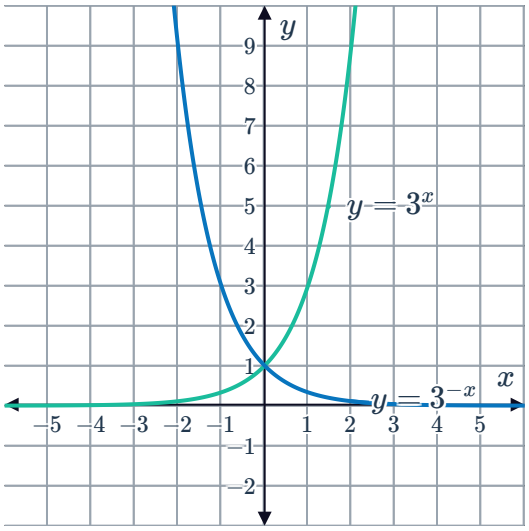
b What is the y-value of the shared y-intercept?

c Describe the nature of these functions for large values of x.

23 Consider the graph of the following functions $y = 3^x$ and $y = 3^{-x}$:

a State the coordinates of the point of intersection of the two curves.

b Describe the behavior of both these functions for large values of x.



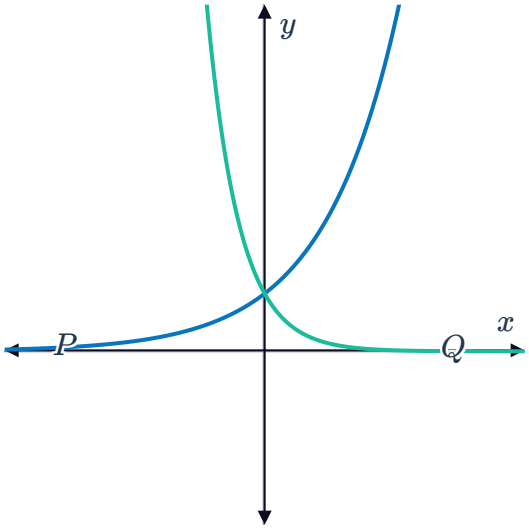
24 Consider the given graphs of the two exponential functions P and Q:
State whether the following pairs of equations could be the equations of the graphs P and Q:

a $P : y = 2^x$
$Q : y = 2^{-x}$

b $P : y = (3.5)^x$
$Q : y = 6^{-x}$

c $P : y = 2^x$
$Q : y = 5^{-x}$

d $P : y = 5^x$
$Q : y = 2^{-x}$

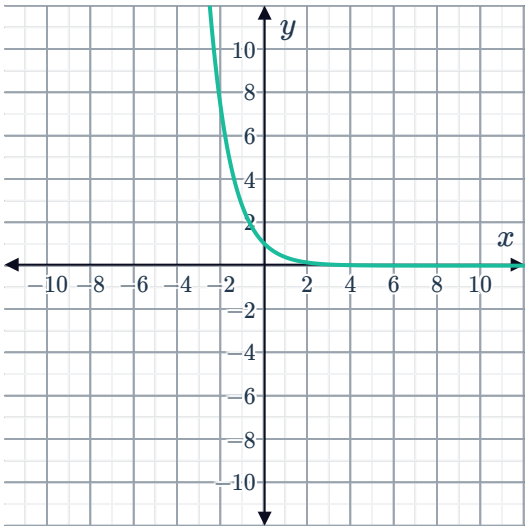


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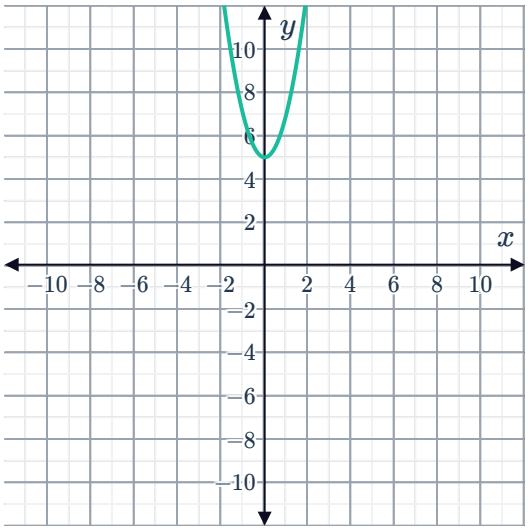
Graphs of exponential functions

25 Determine whether the following graphs show an exponential function:

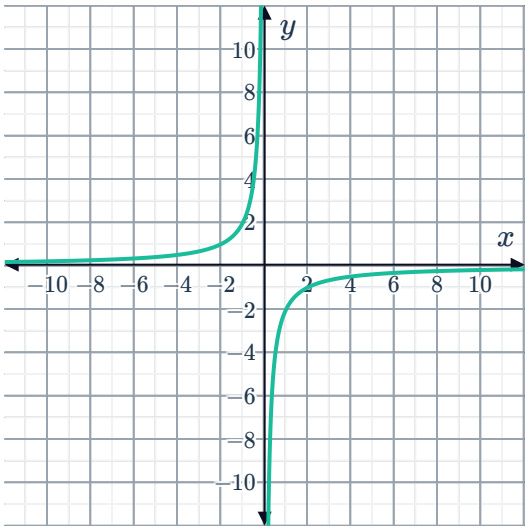
a



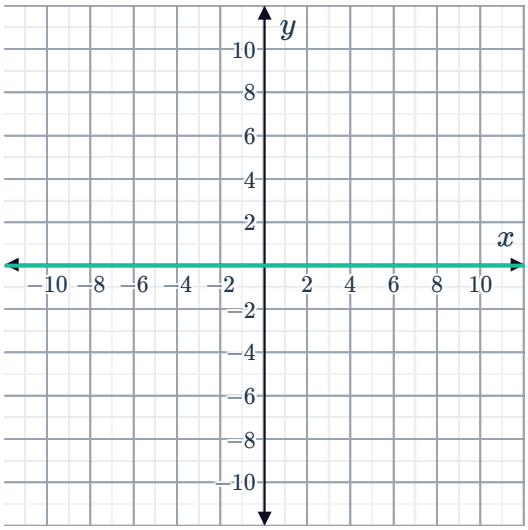
b



c

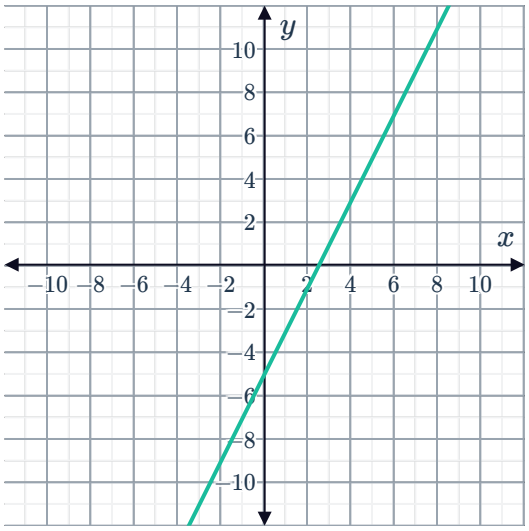


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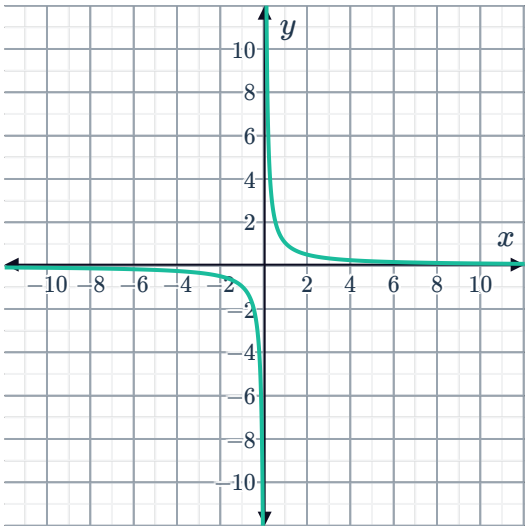


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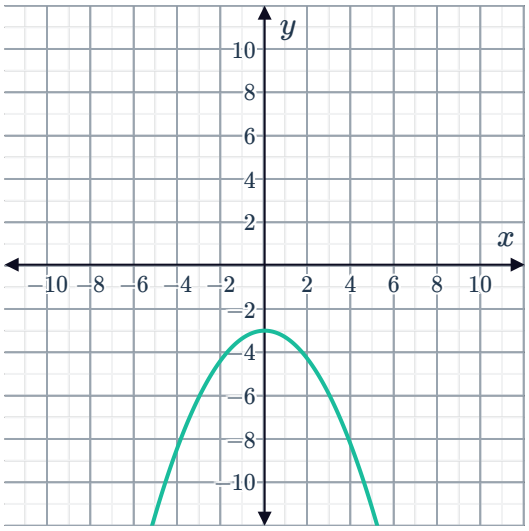
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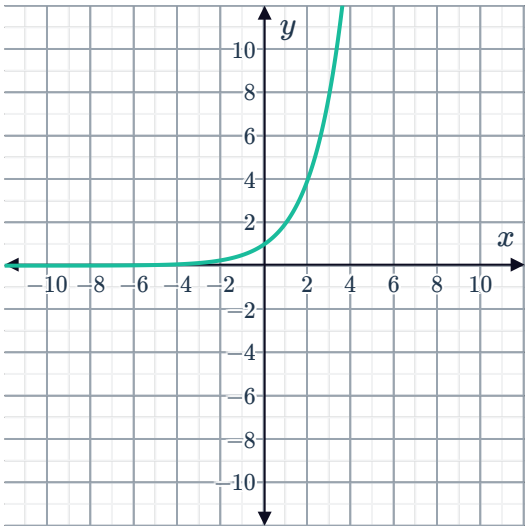
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g



h



26 Consider the equation $y = 3^x$.

a Complete the table of solutions for the given equation.

b Plot the points C, D, and E on a coordinate plane.

c Sketch the curve that results from the entire set of solutions for the equation being graphed.

Point	Coordinates
A	$(-3, \square)$
B	$(-2, \square)$
C	$(-1, \square)$
D	$(\square, 1)$
E	$(1, \square)$
F	$(2, \square)$
G	$(3, \square)$

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27 Consider the equation $y = 2^x$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Sketch the graph of the function.

c What happens to the y-values as x increases?

d What happens to the y-values as x decreases?

e Does the curve crosses the x-axis?

f At what value of y does the graph cross the y-axis?

28 Consider the equation $y = 2^{-x}$.

a Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

b Sketch the graph of the function.

c What happens to the y-values as x increases?

d What happens to the y-values as x decreases?

e Does the curve crosses the x-axis?

f At what value of y does the graph cross the y-axis?

29 For each of the following equations:

i Find the y-intercept of the curve.

ii Complete the table of values:

x	-3	-2	-1	0	1	2	3
y							

iii Find the horizontal asymptote of the curve.

iv Sketch the graph of the function.

a $y = 6^x$ **b** $y = 3^x$ **c** $y = 10^x$ **d** $y = 3^{-x}$

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34 Consider the function $y = -3^{-x}$.

a Find the y-intercept of the curve $y = -3^{-x}$.

b Find the horizontal asymptote of the curve $y = -3^{-x}$.

c Now, sketch the graph of the function.

d Is the function $y = -3^{-x}$, an increasing or decreasing function?

35 Consider the function $y = \left(\frac{1}{2}\right)^x$.

a Determine whether the following functions are equivalent to $y = \left(\frac{1}{2}\right)^x$:

i $y = \frac{1}{2^x}$ **ii** $y = 2^{-x}$ **iii** $y = -2^x$ **iv** $y = -2^{-x}$

b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{2}\right)^x$ from the graph of $y = 2^x$.

c Graph the functions $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ on the same coordinate plane.

36 Consider the function $y = \left(\frac{1}{3}\right)^x$.

a Rewrite the function in the form $y = k^{-x}$.

b Describe a transformation that would obtain the graph of $y = \left(\frac{1}{3}\right)^x$ from the graph of $y = 3^x$.

c Graph the functions $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$ on the same coordinate plane.

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37 Consider the exponentials $y = 3^x$ and $y = 2^x$.

a Find the equation of the horizontal asymptote of the curve $y = 3^x$.

b Find the equation of the horizontal asymptote of the curve $y = 2^x$.

c Complete the table of values for $y = 3^x$:

x	-3	-2	-1	0	1	2	3
y							

d Complete the table of values for $y = 2^x$:

x	-3	-2	-1	0	1	2	3
y							

e Now, sketch the graphs of $y = 3^x$ and $y = 2^x$ on the same coordinate plane.

f State the coordinates of the point at which the two graphs intersect.

38 Consider the functions $y = 2^x$, $y = 3^x$ and $y = 5^x$.

a Sketch the three functions on the same coordinate plane.

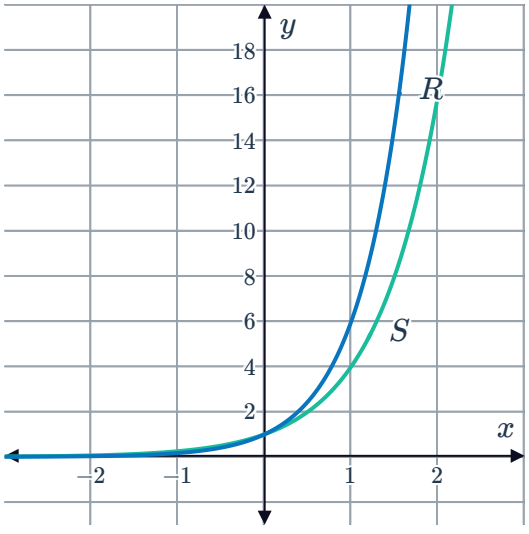
b State the y-intercept of each curve.

c Describe the nature of these functions for increasingly large values of x.

39 Consider the graphs of the two exponential functions R and S:

a One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$.
Which is the graph of $y = 6^x$?

b For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.



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40 Consider the equation $y = -10^x$.

a Jenny thinks she has found a set of solutions for the equation as shown in the table:

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

She notices that all the y-values are negative and concludes that for any value of x, y must always be negative. Is she correct? Explain your answer.

b Sketch the graph of $y = -10^x$.

c Find the values of x for which $y = 0$.

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